Yizhuo Tan

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EDUCATION

Yale University, Connecticut, United States *09/2023 – Present*
Ph.D. in Computer Science *Expected graduation: 2028*
Advisors: Prof. Jakub Szefer (now at Northwestern University) and Prof. Yongshan Ding
Area: Quantum Computer Architecture, Quantum Computing Security

University of Science and Technology of China (USTC), Hefei, China *09/2019 – 07/2023*
Bachelor of Science in Physics
Major: Physics (Atomic and Molecular Physics track)
Minor: Computer Science and Technology

TECHNICAL SKILLS

Programming Languages	Python, C, C++, MATLAB, Verilog, LaTeX
Quantum Computing	Qiskit, pytket, Stim, Qiskit Metal, Amazon Braket SDK, COMSOL, Finite Element Analysis
Software & Tools	Mathematica, Blender, Adobe, ANSYS

RESEARCH EXPERIENCE

Doctoral Research / Research Assistant – Yale University *09/2023 – Present*
Advisors: Prof. Jakub Szefer and Prof. Yongshan Ding

Logical Operation Identification in qLDPC Codes [Code]

- Developed and implemented likelihood-based inference to identify logical operations from syndrome histories in qLDPC codes, using exact fault marginalization and scalable approximations that reduce computation and memory costs.
- Constructed joint likelihood models to identify individual logical instructions (code automorphisms, T-state injection, and logical Pauli measurements) and logical circuits built from these instructions. The models account for unknown faults, measurement randomness, and classical feedforward during circuit execution.
- Analyzed identification limitations and approximation failures. Refined intermediate-state selection to improve identification accuracy from 24.31% to 90.23% on a high-noise benchmark, with only 3.01% higher mean query wall time and 1.83% higher peak process-tree memory usage.

Predictive Reliability Analysis of Superconducting Quantum Processors (EPAR)

- Co-developed EPAR, a framework that predicts layout-dependent robustness of superconducting processors by linking electromagnetic modeling, effective Hamiltonians, and control-pulse response.
- Constructed superconducting circuit models and chip layouts from target Hamiltonians and contributed to layout implementation in Qiskit Metal.
- The framework quantifies deviations between intended and effective qubit connectivity and characterizes the sensitivity of parasitic couplings to local parameter variations.

Private Delegated Quantum Queries (OQRAM)

- Co-designed oblivious QRAM protocols for clients with limited quantum resources, protecting delegated coherent queries through encrypted and shuffled database layouts and quantum-secure address masking.

– Co-developed security and resource analyses for quantum-secure pseudorandom-permutation and quantum one-time-pad variants. Established online quantum communication linear in the combined address and ciphertext register widths and quantified classical database refresh costs.

Pulse-Level Security of Superconducting Processors (QubitHammer)

– Designed and validated pulse-level attacks on IBM and Rigetti superconducting processors under a multi-tenant execution model. Demonstrated state changes across disjoint qubit allocations, including non-adjacent victim qubits, using public control interfaces without elevated privileges.
– Demonstrated that the attacks could bypass the evaluated dynamical-decoupling and active-padding configurations in hardware experiments.
– Implemented Rigetti experiment workflows in Python with Amazon Braket, covering calibration retrieval, hardware execution, and measurement analysis. Released a public demonstration notebook.

Readout Crosstalk in Cloud Quantum Computers

– Characterized correlations among qubit readout outcomes to quantify readout crosstalk and information leakage on superconducting quantum hardware.

Research Visit – Northwestern University *07/2026 – 09/2026*

Host: Prof. Jakub Szefer

– Research on logical-operation identification in qLDPC codes as part of doctoral research.

Undergraduate Research Assistant – USTC (Hefei, China) *09/2021 – 03/2022*

Microscale Nanoscience Laboratory

– Built and aligned optical paths and reflectors for Gaussian boson sampling experiments.
– Assisted in developing and arranging retarder-based optical measurements.

Undergraduate Experimental Project – USTC (Hefei, China) *09/2021 – 12/2021*

Awarded: First Prize, 17th Physical Innovation Research Experimental Paper Competition, USTC *2021*

– Designed ultrasonic phased-array hardware with independently controlled channel phase and drive amplitude for acoustic levitation.
– Modeled acoustic radiation potentials in MATLAB, evaluated fields and particle trajectories in COMSOL, and demonstrated stable levitation and controlled motion.

PUBLICATIONS & PREPRINTS

Peer-reviewed Publications

Navnil Choudhury, **Yizhuo Tan**, Jiaqi Yu, Jakub Szefer, Kanad Basu. "EPAR: Electromagnetic Pathways to Architectural Reliability in Quantum Processors" in Proceedings of the Chips to Systems Conference DAC, 2026. [DOI]

Yizhuo Tan, Navnil Choudhury, Kanad Basu, Jakub Szefer. "QubitHammer: Remotely Inducing Qubit State Change on Superconducting Quantum Computers" in Proceedings of the IEEE International Symposium on Hardware Oriented Security and Trust, HOST, 2026, pp. 151–161. [DOI]

Yizhuo Tan, Jakub Szefer. "I Know What You Are Reading: Evaluating Readout Crosstalk in Cloud-based Quantum Computers" in Proceedings of ACM QSec: Quantum Security and Privacy Workshop @ ACM Conference on Computer and Communications Security, QSec@CCS, 2025. [DOI]

Yizhuo Tan, Hrvoje Kukina, Jakub Szefer. "Securing HHL Quantum Algorithm against Quantum Computer Attacks" in Proceedings of Hardware and Architectural Support for Security and Privacy @ IEEE/ACM International Symposium on Microarchitecture, HASP@MICRO, 2025. [DOI]

Pranet Sharma, **Yizhuo Tan**, Konstantinos-Nikolaos Papadopoulos, Jakub Szefer. "Evaluation of Noise and Crosstalk in Neutral Atom Quantum Computers" in Proceedings of the Quantum-Classical Orchestration for Resilient Engineering Workshop (Q-CORE) at the IEEE International Conference on Quantum Computing and Engineering (QCE), 2025. [DOI]

Yizhuo Tan, Hrvoje Kukina, Jakub Szefer. "Study of Attacks on the HHL Quantum Algorithm" in Proceedings of IEEE Workshop on Quantum IntelLigence, Learning & Security@ IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications, QUILLS@TPS-ISA, 2024. [DOI]

Yizhuo Tan, Chuanqi Xu, Jakub Szefer. "Exploration of Timing and Higher-Energy Attacks on Quantum Random Access Memory" in Proceedings of the International Conference on Computer-Aided Design, ICCAD, 2024. [DOI]

Preprints & Manuscripts

Yizhuo Tan, Shifan Xu, Yongshan Ding, Jakub Szefer. "Identifying Logical Operations from Syndrome Histories for qLDPC Codes." arXiv:2610.40052, 2026.

Shifan Xu*, **Yizhuo Tan***, Yongshan Ding, Jakub Szefer. "OQRAM: Oblivious Quantum Random Access Memory for Securing Delegated Quantum Queries." arXiv:2607.27171, 2026. (* Equal contribution.)

TEACHING EXPERIENCE

Teaching Assistant – Yale University

General Physics Laboratory	*Fall 2026*
Introduction to Computer Engineering	*Spring 2025*
Introduction to Electronics	*Fall 2024*

Undergraduate Teaching Assistant – USTC (Hefei, China)

Mechanics B	*Fall 2022*

HONORS & AWARDS

Best Paper, QSec@CCS	*2025*
Best Poster Award – 1st Place, New England Hardware Security Day	*2025*
Silver Award of Excellent Student Scholarship, USTC (Top 2%)	*2022, 2021*
First Prize in 17th Physical Innovation Research Experimental Paper Competition, USTC	*2021*
Scholarship of Yan Jici Talent Program in Physics, USTC (Top 10%)	*2021, 2020*
Second Prize in 6th China Undergraduate Physics Experiment Competition	*2020*
Chen Linyi Scholarship, USTC (Top 1%)	*2020*
Provincial First Prize in 35th Chinese Physics Olympiad	*2018*

SELECTED PRESENTATIONS

IEEE International Symposium on Hardware Oriented Security and Trust (HOST), McLean, VA	*2026*
HASP@58th IEEE/ACM International Symposium on Microarchitecture (MICRO), Seoul, Korea	*2025*
ACM QSec@ACM Conference on Computer and Communications Security (CCS), Taipei, Taiwan	*2025*
Joint March Meeting and April Meeting - Global Physics Summit, Anaheim, CA	*2025*
IEEE Workshop on Quantum IntelLigence, Learning & Security (QUILLS), Washington, DC	*2024*
ACM/IEEE International Conference on Computer-Aided Design (ICCAD), Newark, NJ	*2024*
2nd Quantum Computer Cybersecurity Symposium, New Haven, CT	*2024*